

Ethernet Interface for Automation Systems

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Communication with remotely located automation systems is possible via the Internet. A physical communication port with a network is required for this communication. An Ethernet interface can be used for that purpose.

Presented here is the design and implementation of an Ethernet interface for automation systems, using a microcontroller. Ethernet is

quite a complex interface, which was difficult to use with small microcontrollers having little memory—until Microchip came up with ENC28J60 Ethernet chip. It is a small chip with only 28 pins that can be used as an Ethernet network interface for any microcontroller equipped with serial peripheral interface (SPI).

Therefore Microchip opens a whole new world of applications like the one we have implemented in this project where real-time readings of tempera-

ture and ambient light sensors placed at a remote place can be read using the Internet.

Fig. 1 shows the html page embedded in the source code of the automation system that provides these readings. The html page can be accessed from a remote location. Distance is no longer a limiting factor. Even Wi-Fi connectivity is possible because the devices can be connected to a wireless bridge too.

Circuit and working

Fig. 2 shows the block diagram of the overall automation system. The temperature and light intensity are continuously monitored by two sensors that communicate with the microcontroller unit via I²C protocol. The relay is activated based on the readings from the ambient light sensor. Real-time reading from the sensors and the trigger status, for example the status of temperature, are continuously displayed on the LCD.

The automation system is connected to the router using Ethernet interface. The html page embedded in the code is continuously updated with real-time readings of both the sensors as shown in Fig. 1.

Fig. 3 shows the circuit built around microcontroller ATmega128 (IC3), Ethernet controller ENC28J60 (IC4), regulators 7805 and LM1117 (IC1 and IC2), digital-out temperature sensor TMP275 (IC5), miniature ambient light photo-sensor APDS9300 (IC6) and a few discrete components.

Ethernet interface. The ENC28J60 is a standalone Ethernet controller with an industry-standard SPI interface. It is designed to serve as an Ethernet network interface for any controller equipped with SPI. The ENC28J60



Fig. 1: html page embedded in the code

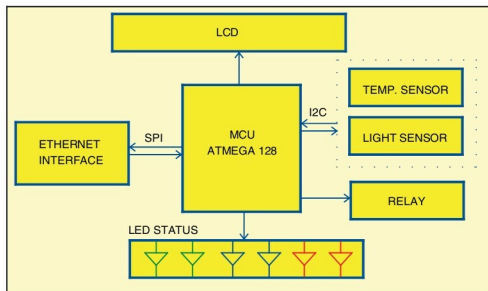


Fig. 2: Block diagram of automation system